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Intelligent News Summarization & Categorization Agent

From Design to Functional Implementation

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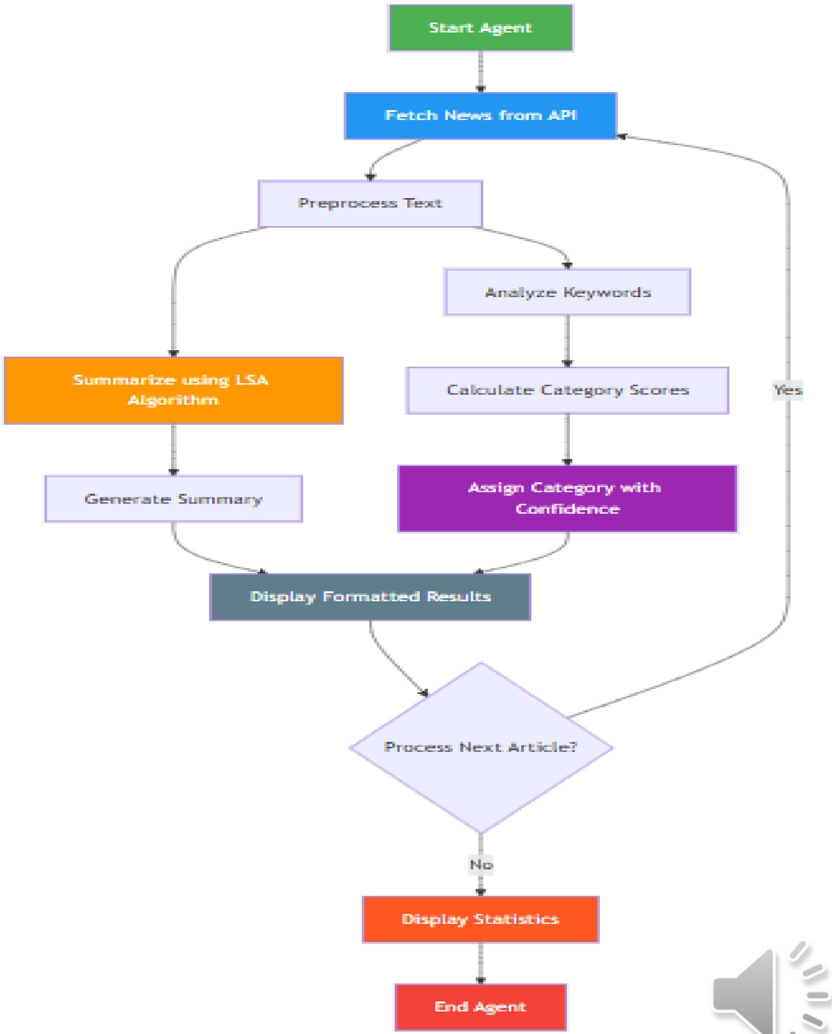
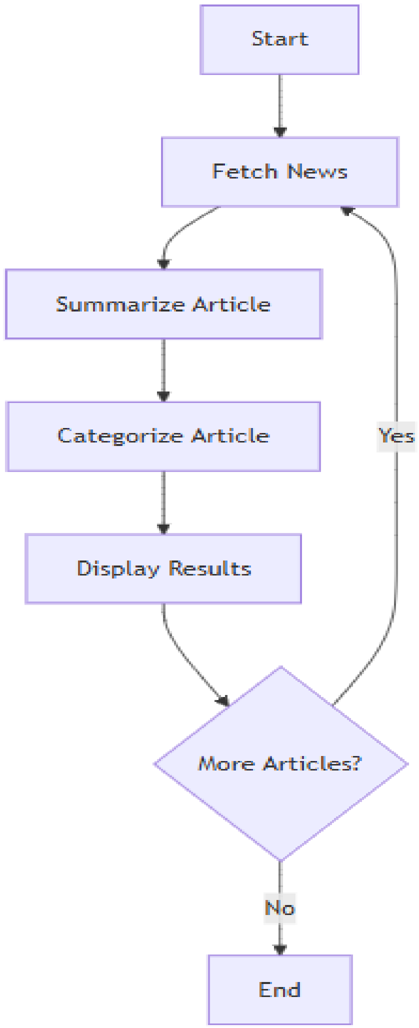
Project Overview & Objectives

- **Problem:** Information overload. Too many news articles, not enough time.
- **Solution:** An autonomous agent that curates and digests news.
- **Core Objectives:**
  - Fetch real-time news from a reliable source (NewsAPI).
  - Generate concise summaries using Natural Language Processing (NLP).
  - Classify articles into relevant categories.
  - Present a clean, user-friendly output.



System Architecture & Design

- **Input:** NewsAPI (Cloud Service)
- **Core Agent:**
  - Data Fetcher Module
  - Text Preprocessor
  - Summarization Engine
  - Categorization Engine
- **Output:** Console/File Report
- **Key Points:**
  - Modular design for easy testing and maintenance.
  - Each component has a single, clear responsibility.
  - Follows a sequential pipeline architecture common in intelligent agents.



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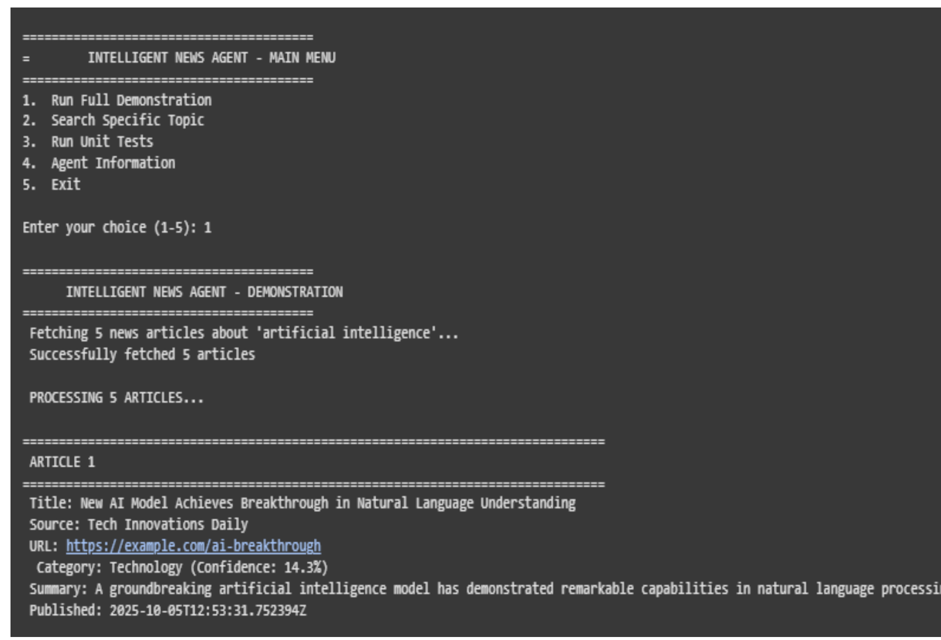
Implementation Deep Dive - Data & Summarization

- **Data Fetching:** Using the requests library to call the NewsAPI. The agent authenticates with an API key.

- **Summarization:** Implemented using the sumy library's LsaSummarizer. This is an **unsupervised learning** algorithm (Latent Semantic Analysis) that identifies the most significant sentences in a text. We chose this

- for its efficiency and lack of need

- for pre-trained models.



Implementation Deep Dive - Categorization

- **Technique:** A lightweight, rule-based classifier using keyword matching.
- **Why this approach?** For a well-defined set of categories (Tech, Business, Health), it's fast, transparent, and requires no training data. It's a pragmatic choice for an agent with a focused goal.
- The keywords were curated based on common terms in each domain (e.g., 'python', 'AI', 'software' for Technology).

```
==========================================================================
ARTICLE 1
==========================================================================
Title: New AI Model Achieves Breakthrough in Natural Language Understanding
Source: Tech Innovations Daily
URL: https://example.com/ai-breakthrough
 Category: Technology (Confidence: 14.3%)
Summary: A groundbreaking artificial intelligence model has demonstrated remarkable capabilities in natural lang
Published: 2025-10-05T12:53:31.752394Z

==========================================================================
ARTICLE 2
==========================================================================
Title: Python 4.0 Release Announcement Expected Next Month
Source: Developer Weekly
URL: https://example.com/python-4-release
 Category: Technology (Confidence: 17.9%)
Summary: After years of development, Python 4.0 is set to be unveiled next month with significant performance en
Published: 2025-09-22T12:53:31.752417Z

==========================================================================
ARTICLE 3
==========================================================================
Title: Global Stock Markets Reach Record High Amid Economic Recovery
Source: Financial Times
URL: https://example.com/stock-market-high
 Category: Business (Confidence: 21.4%)
Summary: Global financial markets are experiencing unprecedented growth with major indices breaking historical r
Published: 2025-09-28T12:53:31.752423Z
```

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Demonstration of the Agent in Action

- "As you can see, the agent successfully fetched 5 articles."
- "Let's look at this one about a new iPhone. The summary accurately captures the main point, and it has been correctly categorized as 'Technology' based on the keyword 'iPhone'."
- "This demonstrates the core functionality working as designed."

```
                   PROCESSING STATISTICS
========================================
 Total Articles Processed: 5

 Category Distribution:
   • Technology: 3 articles (60.0%)
   • Business: 1 articles (20.0%)
   • Health: 1 articles (20.0%)

 Demonstration completed successfully!
 This demo used mock data to showcase full agent functionality

Press Enter to continue...

========================================
=        INTELLIGENT NEWS AGENT - MAIN MENU
========================================
1.  Run Full Demonstration
2.  Search Specific Topic
3.  Run Unit Tests
4.  Agent Information
5.  Exit

Enter your choice (1-5): 2
Enter search topic: 3
Number of articles (1-10): 4
```

Testing & Validation Strategy

- **Unit Testing (with** unittest**):**
  - Tested the categorizer with known inputs (e.g., "Microsoft profits soar" -> "Business").
  - Tested the text preprocessing function.
- **Functional Testing:**
  - Ran the full agent pipeline with different search queries.
  - Verified output format and completeness.
- **Remediation:** Initially, some summaries were too long. Adjusted the sentence count parameter in sumy from 5 to 3 to improve conciseness.

Challenges & Learnings

- **Challenge 1:** NewsAPI rate limiting. **Solution:** Implemented error handling and respectful timing between calls.
- **Challenge 2:** Summarizer performance on very short articles. **Solution:** Added a check for article length before summarizing to avoid errors.
- **Key Learning:** The importance of choosing the right tool for the task. A simple keyword categorizer was more effective and efficient than a complex ML model for this specific scope.

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Conclusion & Future Enhancements

- **Conclusion:** Successfully developed a functional intelligent agent that automates news digestion, meeting all design requirements.

- **Future Work:**
  - Integrate a Machine Learning-based classifier (e.g., with scikit-learn) for better accuracy.
  - Add a GUI using tkinter or a web interface with Flask.
  - Implement sentiment analysis to gauge article tone.
  - Allow user preferences for sources and categories.

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Q and A

Free discussion

Thank you